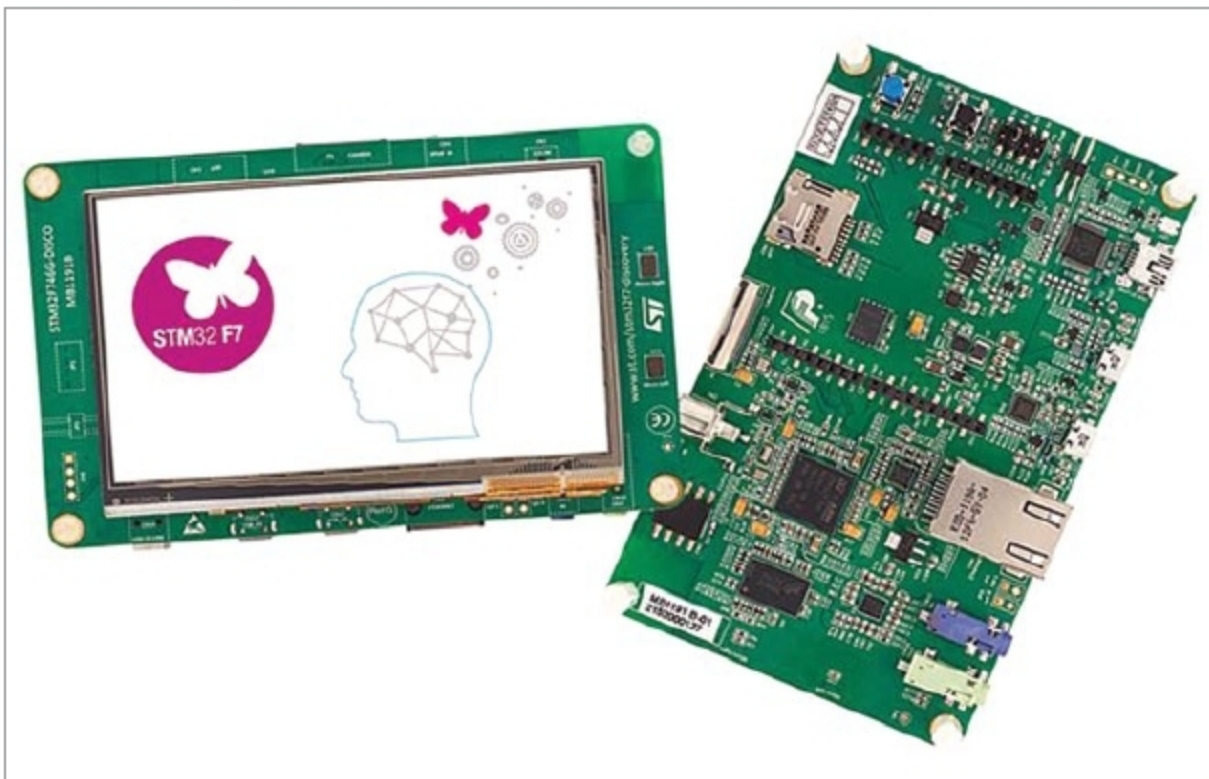




STM32F7, an ARM Cortex-M7 (Credit: www.st.com)

and applications have automated several small tasks like setting up the grid, rules and scales. With fewer tasks at hand, designers can now focus more on the design and decide the best-fit design elements than on the arrangement of those elements. This helps them carry out several tasks in the set time frame, that too with accuracy and perfection.

Designers of the future will be more creatively, technologically and intellectually advanced—all they will need to do is counter-design issues and problems. Almost half of the designing tasks will be automated through AI, making systems more robust and efficient.

AI can detect and analyse patterns and trends on its own and optimise the design system accordingly. Designing apps and tools are much more intelligent than these were ever before. AI-powered tools can also determine the best filter and visual effect to apply to images.

Jacky Wan, vice president of engineering for Arrow's components business in Asia-Pacific region, says, "We are collaborating with Thundersoft to assist developers and engineers to integrate AI technology into Edge/gateway devices. This will empower developers and manufacturers to train and verify algorithms, and create AI applications quickly. It can also assist them in building an optimal intelligent system that can handle speech recognition, complex

event/data processing and computer vision features on Edge/gateway devices."

Latest design developments in the industry

AI is being adopted globally by all industries to drive efficiency, improve productivity and reduce costs. AI-driven systems and applications are making railway operations safer, smarter and more reliable, significantly enhancing passenger travel experience and freight logistics services. These AI-driven applications only function with proper data input that is collected by a large number of IoT devices and cameras installed in stations, on trains and along tracks. Target applications include, but are not limited to, passenger information systems, railroad intrusion detection, train station surveillance, onboard video security and railroad hazard detection.

Crystal Tseng, senior product manager, ADLINK, says, "We believe that AI can transform traditional rail operations by driving improvements including fast and convenient ticket-free check-in, real-time tracking of health diagnostics, personalised infotainment and onboard services, accurate arrival time predictions and rapid response in an emergency." ADLINK's PIS-5500 intelligent platform is installed on special rail inspection trains to process captured images of key wayside equipment in real time.

The application can not only detect suspicious behaviour and trigger alerts but also conduct post-event analysis.

Smart cameras will play a big role in smart cities and connected industrial facilities, and increasingly large streams of data will need to be sent over networks as a result. This could result in slower transmissions, which would limit the functionality of connected devices. Bringing processing and analytics onto cameras allow them to process in real time without having to wait for data to be sent between devices and the server.

Qualcomm has released its next-generation AI-powered camera design platform using its latest purpose-built systems-on-chip (SoC) to better enable on-device ML and processing capabilities. The platform is designed for Edge computing use-cases that need support for video processing and analytics. The SoC can be applied to an array of devices including smart security cameras, smart displays, sports cameras, robotics, inventory management tools, wearable and body cameras, and dash cams.

Smart cameras that contain MediaTek's Helio P90 SoC support faster, complex and more dynamic AI experiences such as human pose detection, which can track and analyse body movements. These also support Google Lens, real-time beautification, deep-learning facial detection, artificial reality and mixed reality acceleration, object and scene identification, and other real-time enhancements for photos and videos.

ST's embedded AI technologies make neural networks run on STM32 MCUs, reducing network bandwidth by performing processing on the Edge side.

A massive revolution is coming as smart devices proliferate the world, driven by technology trends like ubiquitous connectivity, cloud computing, AI and low-cost sensors. Existing products, ranging from cameras to audio speakers, are getting smarter and a whole new range of devices is emerging that can make lives more convenient, safer and entertaining. With upgradation in AI powered devices, we will be able to automate most of the tasks in every field. **EFY**